M. Vigneshwar
240701588
CSE

CN - Observation

S.NO	DATE	TITLE	Teacher's Sign
1	21/7/26	Various network command in Linux & windows	
2	20/7/26	STudy of Network cable	
3	4/8/26	Experiment on CISCO PACKet Tracer (simulation TOOL)	
4	11/8/26	Setup and configure a LAN using a switch and Ethernet cable in your lab	
5	18/8/26	Experiment on packet capture Tool wireshark	
6	25/8/26	Error correction at data link layer	
7	11/9/26	Flow control at data link layer	
8	8/9/26	NMAP To discover live hosts using Nmap scan (ARP, ICMP, TCP/UDP) on The platform	
9	13/9/26	Implementation of subnetting in cisco packet Tracer simulator	
10	29/9/26	Internetworking with router in CISCO packet Tracer simulator	
11	29/9/26	Routing at Network layer	
12	30/9/26	End-End communication at Transport layer	
13	30/9/26	Implement your own ping program	
14	30/9/26	write a code using RAW sockets To implement packet sniffing	

Completed 13/10/26

Ex NO: 1
20/7/26

Various Network command in Linux & windows

Aim:

To study of various network commands used in Linux & windows

Basic Networking commands:

1) arp -a : ARP is short form of address resolution protocol, it will show the IP address of your computer along with the IP address and MAC address of your router

Output:-

gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on enp0s3.

2) hostname : This is the simplest of all TCP/IP command. It simply display the name of your computer.

Output:-

fedora

3. Ipconfig /all :-

This command display detailed configuration information about your TCP/IP connection including router, gateway, DNS, DHCP and type of ethernet adapter, in your system

Host Name : HDC042251

Physical Address : 0A-00-27-00-00

DHCP Enable : NO

IPv4 Address : 192.168.56.1

4 nbtstat -a :

This command help solve problem with Net BIOS name resolution (stands for NetBIOS over TCP/IP)

a. (adapter status) list the remote machine's name table give its name

A (Adapter status) List the remote machine name table gives

5. netstat :

It is a command line tool for monitoring network connection both incoming and outgoing as well as viewing routing Table

Proto	Local Address	Foreign address
TCP	127.0.0.1:55816	HDC042251:55817
TCP	127.0.0.1:55817	HDC042251:55816
TCP	127.0.0.1:58879	HDC042251:55818
TCP	172.16.10.180:7680	172.16.10.137:59679

6 NSlookup :

IT is used To display DNS lookup in linux, such as The IP address of a particular computer

Default server : unknown
Address : 172.16.8.1

7 Pathping :-

Pathping is unique To windows's and is basically a combination of The ping and Tracert commands

-g host-list :- Loose source route along host list

-h maximum_hops Maximum number of hops To search for target

8. Ping :-

Packet Internet Groper) command is The best way To Test connectivity between Two nodes. Ping use ICMP (Internet control message).

64 Bytes from fedora fe80:96 ff:454d:ce50): icmp_seq=1

$H_1 = 64$ time = 0.88ms

PING 04.21.4.8 (34.21:4:8) 56(84) bytes of data

~~Ex No: 2~~

~~Date: 27.6.25~~

Important Linux networking commands

1. IP:

Every administrator will need in daily work, setting up new system

link (lookup 00:00:00:00:00 brd 00:00:00:00:00:00)

inet 127.0.0.1.18 scope host lo

2. If config:

command was a staple in many sysadmin's tool belt for configuration and Troubleshooting networking.

3. mtr

server as a network diagonstic tool

mtr google.com

Packets :

Loss snt cost avg boot wds

0.0% 17.8 0.1 0.1 0.0 0%

4. Tcp dump :-

Designed for capturing and display packet

output :-

Tcp dump -p

Observation :-

1. which command is used to find the reachability of host machine from your device?

Ans: Ping

Eg: ping google.com

2. which command gives the details of hopes taken by a packet to reach its destination?

Ans:- mtr, It show route from a computer to a specified host mtr provides a lot of statistics about each hop such as response time and percentage

3. which commands displays the ip configuration of your machine

⇒ ip command can show address information manipulate routing plus display network various devices interface & funnels.

4. which command display the TCP port states in your machine?

=> net stat

It display variety of statistic about a comp active TCP/IP connection

5. write the modify the ip configuration in a linux machine.

=> Assigning IP address to interfaces ip address add 192.168.1.254/24.

* Deleting IP address

ip address del 192.168.1.254/24 dev ens

Ex:2 Study of Network cables

Aim:

To study different type and Network cables and make a cross-wired cable and straight Through cable using clamping / crimping tool.

a) understand different types of network cable.

1. unshielded Twisted Pair (UTP) cable
2. shielded Twisted Pair (STP) cable
3. coaxial cable
4. Fiber optic cable

cable Type	category	Maximum Data Transmission	Advantage / Disadvantage	use	Image
UTP	category 3 category 5 category 5e	10 bps up to 100 mbps 1 Gbps	Ad: * cheap in cost * Easy to install Disad: more prone to (EMI)	10 base T ethernet fast ethernet gigabit ethernet	
STP	category 6, 6a	10 Gbps	Ad: * shielded * Faster than UTP * less susceptible to noise and interference	gigabit ethernet 10G ethernet (55m) widely used in data center.	

SSTP	Category 7	10 Gbps	disad :- * expensive * greater installation effort	gigabit ethernet, 10G ethernet
coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	Hi Gh bandwidth * Low loss Bandwidth disad :- • Limited distance * cost * size is bulky	speed of signal is 500m Television network high speed internet connection
Fiber optics cable	single mode multi mode	100 Gbps	Ad:- * high speed * high Bandwidth * high security disad: * Expensive * Required skilled installed	* maximum distance of fiber optical cable in around 100 meter

b) Make your own ethernet cross-over cable / straight cable.

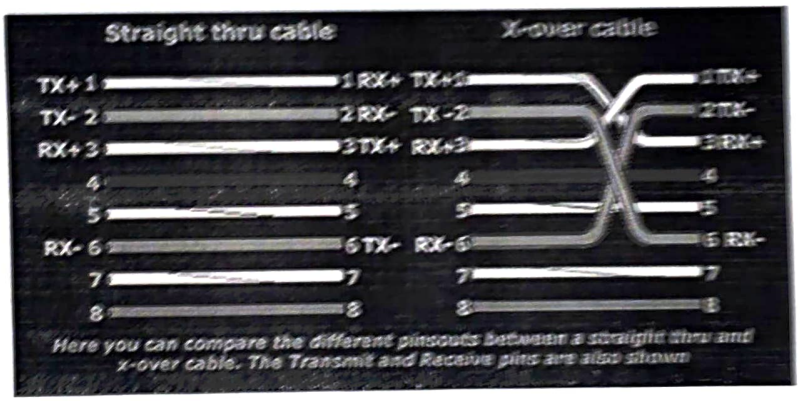
Tools and parts needed:

* ethernet cabling - can be a certified for gigabit support, but CAT 5 cabling could be used for over shorter distances.

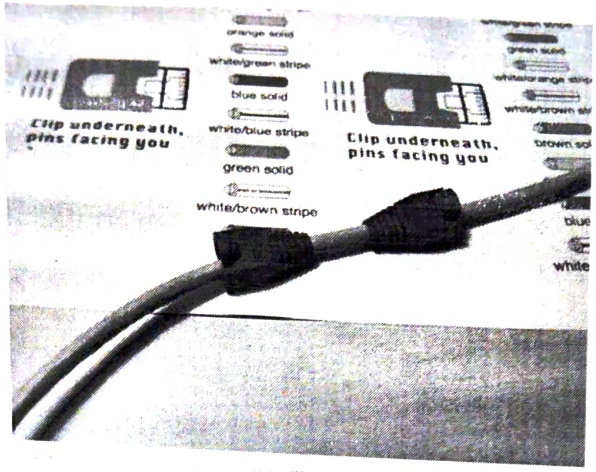
* A crimping tool. There is an all-in-one networking tool created to push down the pins in the plug and strip and cut the shielding off the cables

* Two RJ45 plugs.

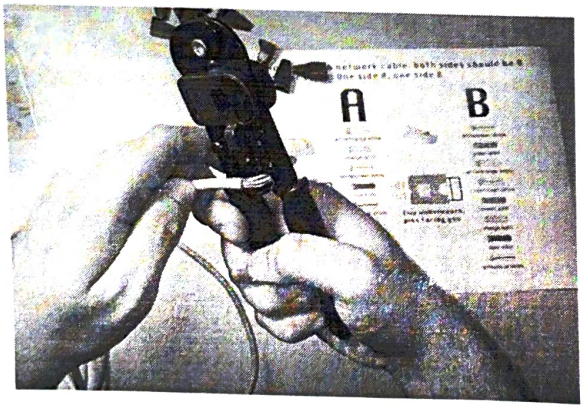
* optional two plug shields.



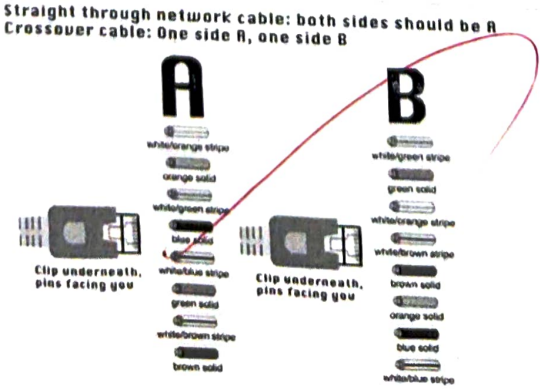
Step 1: To start construction of the device, begin by threading shields onto the cable.



Step 2: Next, strip ~~this~~ approximately 1.5 of cable shielding from ends, the crimping tools has a round area to compute this task.

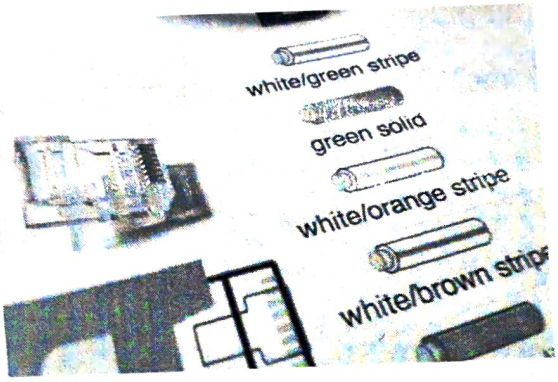


Step 3: After, you will need to untangle the there should be four "Twisted pairs", referring back to the sheet, arrange them from top to bottom.



Step 4:

once the order is correct, bunch them together in a line and if there are any that stick out farther than others, skip them back to create an even level, the difficult aspect is placing these into the RJ45 plug without messing up the order.



Step 5: Next, push the cable right in the notch at the end of the plug needs to be just over the cable shielding and if it isn't that means that you stripped of too much shielding. simply snip the cables backs a little more.



Step:6 After the wires as securely sitting inside the plug, insert in into the crimping tool and push down.

It should be shaped correctly, but pushing too hard can crack the fragile plastic plug.

Observation :-

Straight cable	cross cable
same wiring standard on both ends	Different wiring standard on each end
used to connect different Types of devices	used to connect similar Type of devices
Eg:- Pc to switch	Eg: Pc to Pc

2. which Type of cable is used to connect Two Pcs?

⇒ cross cable because both Pcs are similar devices, so the Transmit (Tx) and receive (Rx) wires must be crossed to communicate directly

3. Which Type of cable is used To connect a router/ switch To your PC ?

⇒ Straight cable, because a PC and a switch / router are different Type of networking devices

4. Find out The category of Twisted pair cable used in your lab To connect The PC To The network socket.

⇒ category : cat 5e on cat 6.

5. write down your understanding, challenges faced, and output received while making a Twisted pair cross / straight cable.

⇒ * I learned The difference btw straight & cross cables

* I understood The T568A and T568B wiring standards

* Arranging The wire color in The correct order.

* Tested The cable using a cable tester.

Ex: 3 Experiment on CISCO PACKET TRACER (Simulation TOOL):

Aim:-

To study the Packet Tracer Tool Installation and User Interface overview

Introduction:-

-> It allow you to model complex systems without the need for dedicated equipment.

-> It help you to practice your network configuration and Troubleshooting skills via computer on an ~~Android~~ Android OR IOS based mobile devices

-> It is available for both the Linux and window desktop environments.

User Interface overview:

1. Menu bar - This is a common menu found in all software applications; it is used to open, save, print.

2. Main Toolbar - This bar provides shortcut icons to menu options. That are commonly accessed, such as open, save, zoom, undo and redo and the right hand side is an icon for entering network information.

3. Logical / Physical workspace Tabs - These Tab allow you to toggle btw the logical and Physical work areas.

4. Workspace - This is the area where Topologies are created and simulations are displayed

5. Common tool bar - This Tool bar provides control for manipulating topologies such as select, move layout, place note, delete, inspect and PDU

6. Real-Time / simulation tabs - These Tabs are used To Toggle btw the real and simulation modes.

7. Network component box - This component contain all of the network and end devices available

7.a) Device Type selection box - This area contain devices ~~onto~~ categories area

7.b) Device - Specific selection box - When a devices category is selected, this selection box display The different devices model within That category.

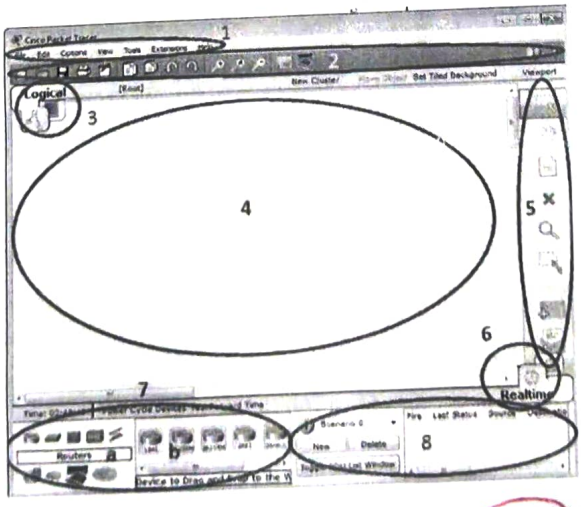
Analyse the behavior of network using CISCO Packet Tracer.

1. From the network component box, click and drag and drop

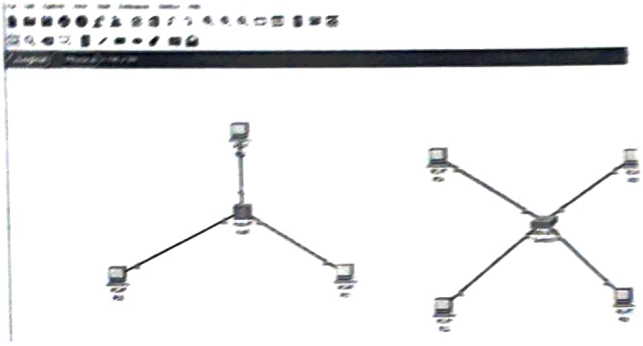
   a. 4 generic PCs and one HUB

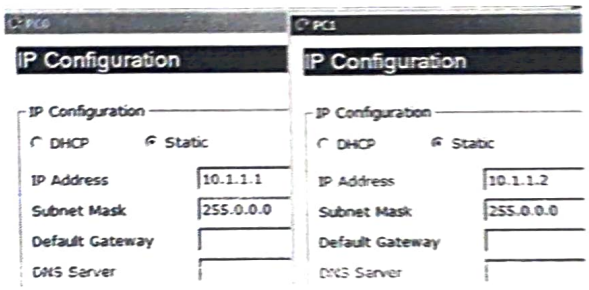
   b. 4 generic PCs and one switch

2. Select one of the PC and connect it to HUB using the cable. The link LED should glow in green, similarly connect remaining 3 PCs to to Hub



3. Click on the PCs connected to hub, go to the Desktop Tab, click on IP configuration and enter an IP address and subnet mask. Here, the default gateway and DNS server information is not needed as there are only two end devices in the network.





4. Observe the flow of PDU from source PC to destination PC by selecting the Realtime mode of simulation

5. Repeat Step-3 to Step-5 for the PCs connected to the switch

Ex NO: 4

Setting up a local Area Network

Aim :-

To design, set up and configure a Local Area network (LAN) connecting four computers through a network switch, assign static IP addresses to all devices, verify connectivity using the ping command.

Equipment Required :-

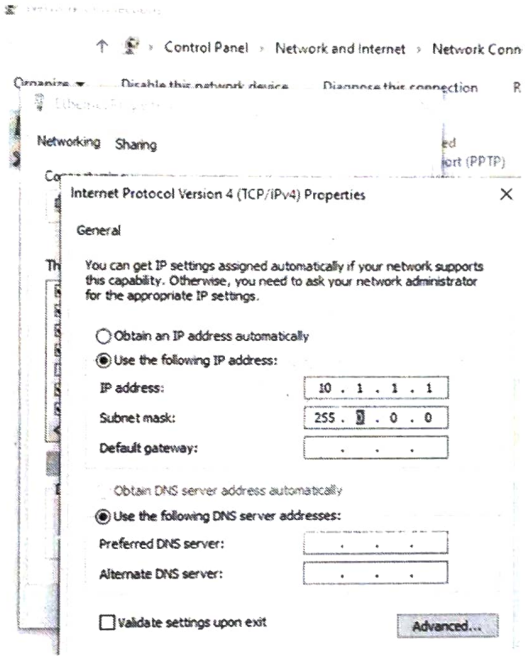
S.NO	Equipment	Quantity	Specification
1.	Personal computer	4	Ethernet (NIC) enabled
2.	Network switch	1	8/16/24 port, unmanaged / managed
3	Ethernet cable	4	Straight-Through, RJ-45 connector.

Theory :-

A Local Area Network (LAN) is a network that connect computer and devices within a limited geographical are such as a home, office on laboratory, allowing them to share resources such as files, printers and internet connection.

Network diagram :-

The LAN was implemented using a star topology as shown below, with 4 PCs connected individually To a central switch.



Implementation into cisco packet tracer :

Since physical hardware was limited The LAN was also implement and verified using cisco packet tracer simulation software following.

Step 1 : place The devices:

→ open cisco packet Tracer

→ From the devices category panel, select End devices and drag 4 PCs onto the work space.

Step: 2 Cable The devices:

-> click connection -> select copper Straight-Through cable

-> click each PC, Then click The switch and select an available Fast Ethernet port To complete Each connection.

Step 3: Assign IP Addresses To PCs

* click a PC -> Desktop tab -> IP configuration

* Enter The static IP Address and subnet mask as per The IP table.

Step 4: configure The switch management IP

* click The switch -> CLI Tab

* Enter The following commands To assign a management IP:

enable
configure Terminal
Interface Vlan 1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit.

Step 5: Verify connectivity using Ping

* Click a PC -> Desktop tab -> command prompt
* use the ping command to test connectivity to the other PC

Observation :-

All four PC were successfully connected to the switch in a star topology and assigned unique static IP address within the same subnet. Ping test btw all devices returned successful replies, confirming end-to-end connectivity. The shared folder / server resource was successfully accessed from other computer on the network.

Ex. No: 5 experiment on packet capture Tool: wireshark

Aim:-

Experiments on Packet capture tool. wireshark

Packet Snifer:-

-> Sniffs messages being sent / received from / by your computer

-> Store and display the contents of the various protocol fields in the messages

-> Passive program

* never send packet itself
* no packets addressed to it
* receives a copy of all packets (sent / received)

Packet sniffer structure Diagnostic tool:-

- Tepdump

-> Eg: tepdum -erx host 10.129.41.2 -w exe3.out

- wireshark

-wireshark -r exe3.out

WIRESHARK:

Wireshark, a network analysis tool formerly known as Ethereal, captures packets in real time and display them in human-readable format.

We can do with wireshark :-

-> capture network Traffic

-> Decode packet protocols using dissectors

-> Define filters - capture and display

-> watch smart statistics

-> Analyze problem

-> interactively browse that Traffic

wireshark used for :-

-> Network administrators : Troubleshoot network problem

-> Network security engineers : examine security problem

-> People : learn network protocol internals

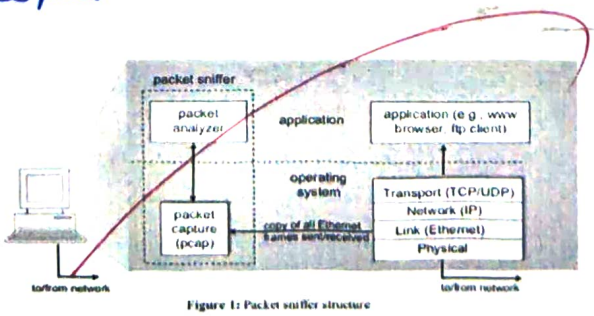
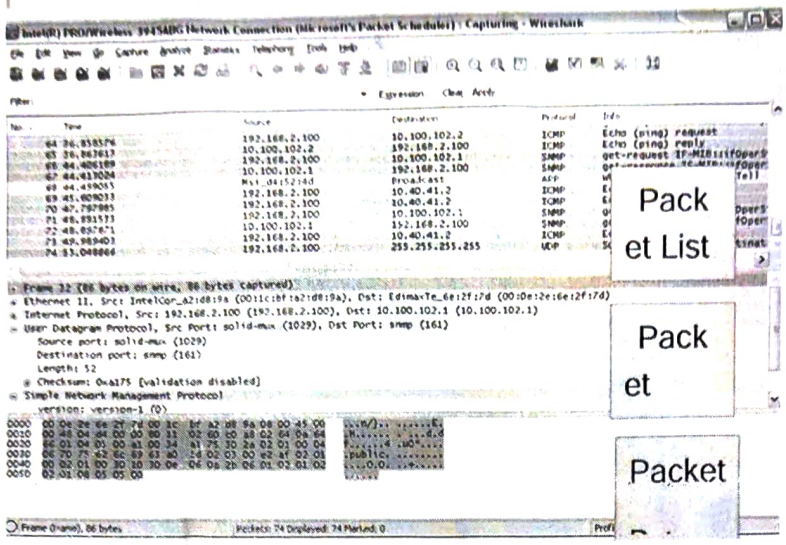
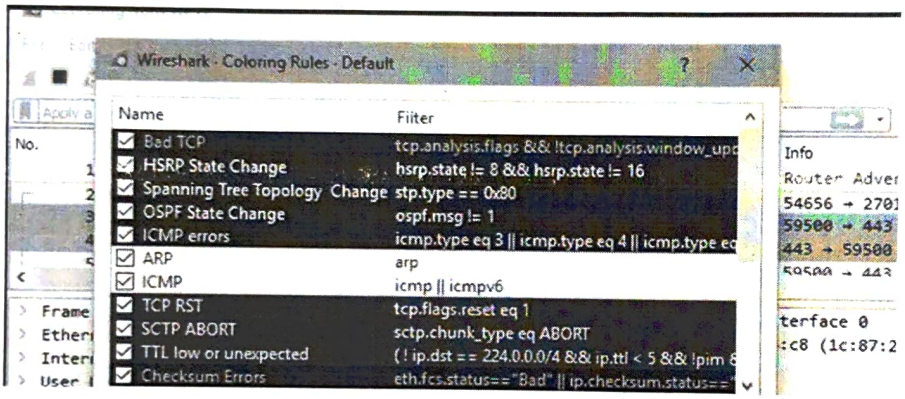
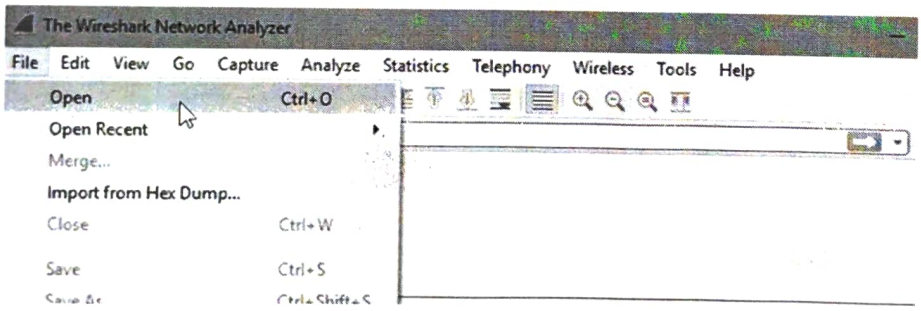
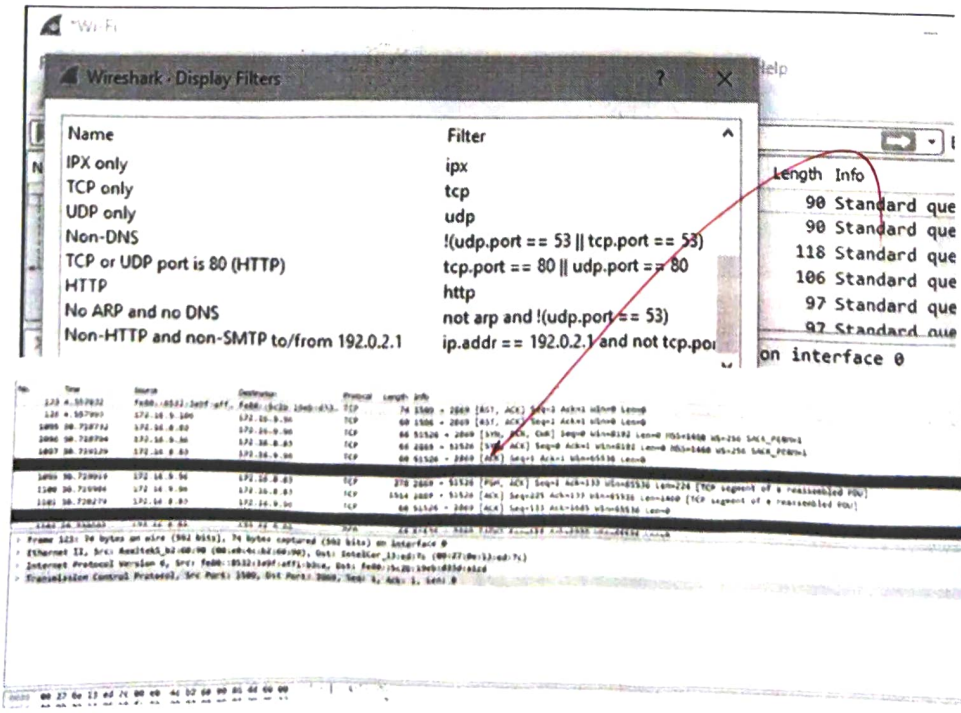


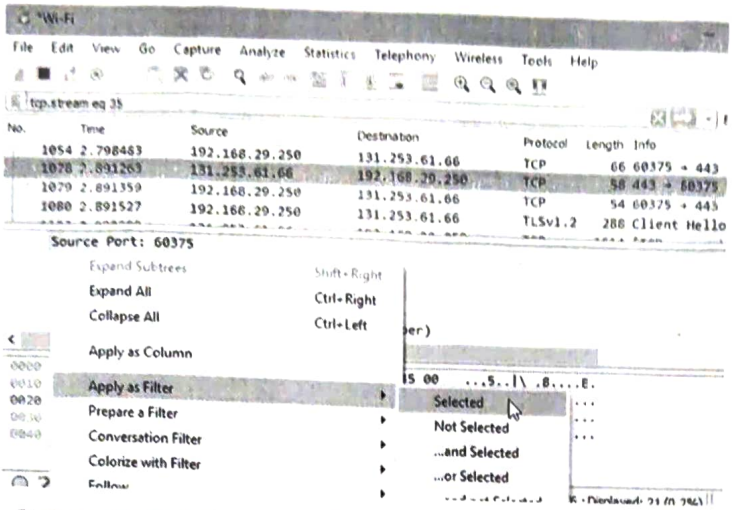
Figure 1: Packet sniffer structure

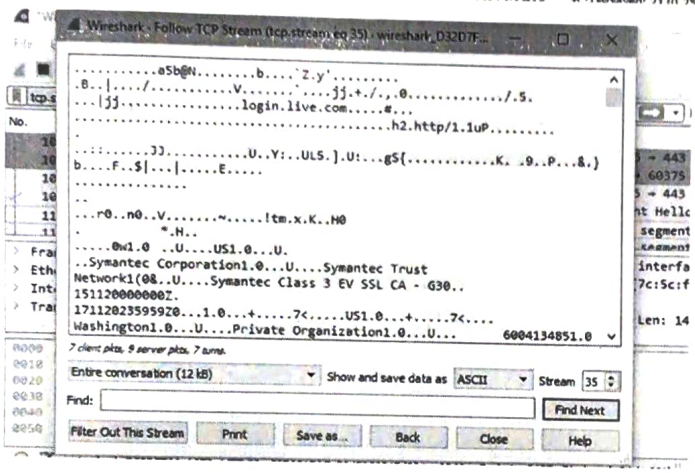

Pack
et List
Pack
et
Packet

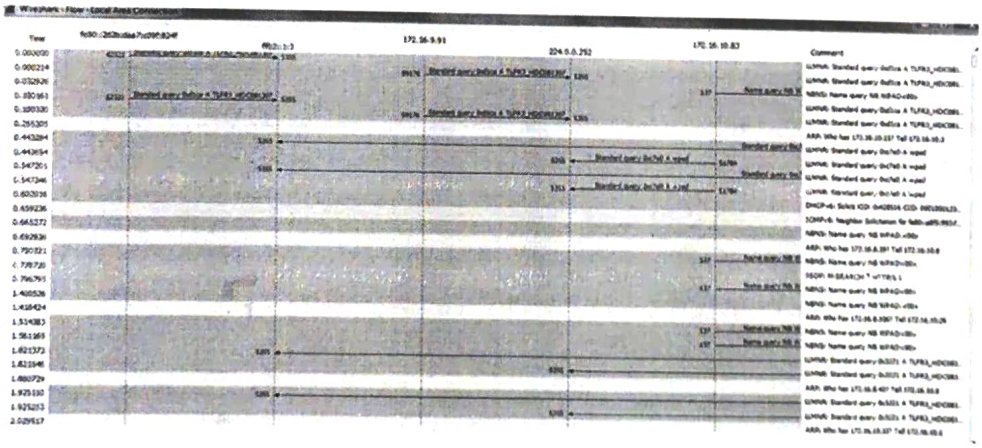

Wireshark · Coloring Rules · Default
Name
Filter
Bad TCP
tcp.analysis.flags && !tcp.analysis.window_upd
HSRP State Change
hsrp.state != 8 && hsrp.state != 16
Spanning Tree Topology Change
stp.type == 0x80
OSPF State Change
ospf.msg != 1
ICMP errors
icmp.type eq 3 || icmp.type eq 4 || icmp.type eq
ARP
arp
ICMP
icmp || icmpv6
TCP RST
tcp.flags.reset eq 1
SCTP ABORT
sctp.chunk_type eq ABORT
TTL low or unexpected
(ip.dst == 224.0.0.0/4 && ip.ttl < 5 && !pim
Checksum Errors
eth.fcs.status=="Bad" || ip.checksum.status==
Info
Router Adver
54656 → 2701
59500 → 443
443 → 59500
terface 0

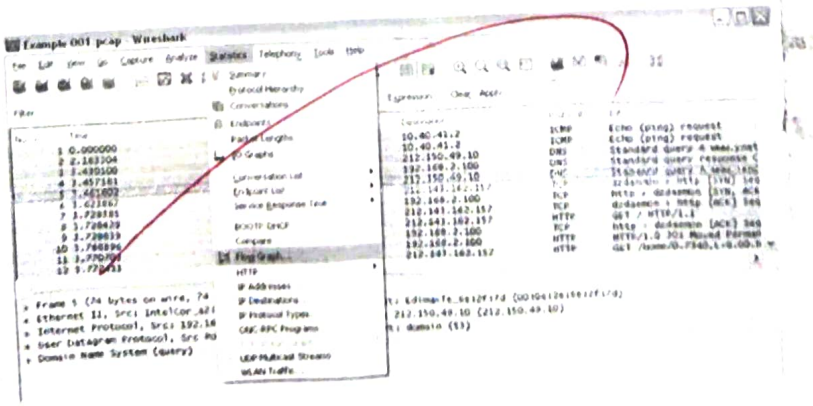

The Wireshark Network Analyzer
File
Edit
View
Go
Capture
Analyze
Statistics
Telephony
Wireless
Tools
Help
Open
Ctrl+O
Open Recent
Merge...
Import from Hex Dump...
Close
Ctrl+W
Save
Ctrl+S

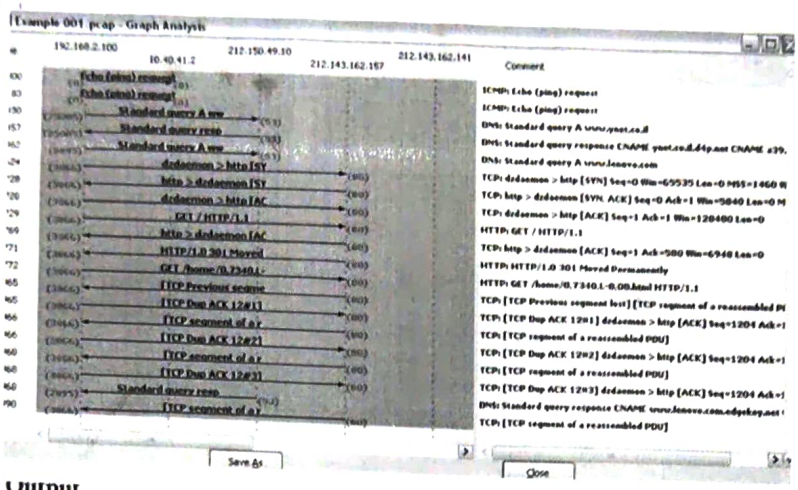

Wireshark · Display Filters
Name
Filter
IPX only
ipx
TCP only
tcp
UDP only
udp
Non-DNS
!(udp.port == 53 || tcp.port == 53)
TCP or UDP port is 80 (HTTP)
tcp.port == 80 || udp.port == 80
HTTP
http
No ARP and no DNS
not arp and !(udp.port == 53)
Non-HTTP and non-SMTP to/from 192.0.2.1
ip.addr == 192.0.2.1 and not tcp.por
Length Info
90 Standard que
90 Standard que
118 Standard que
106 Standard que
97 Standard que
on interface 0


*Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.stream eq 35
No. Time Source Destination Protocol Length Info
1054 2.798483 192.168.29.250 131.253.61.66 TCP 66 60375 → 443
1078 2.891263 131.253.61.66 192.168.29.250 TCP 58 443 → 60375
1079 2.891359 192.168.29.250 131.253.61.66 TCP 54 60375 → 443
1080 2.891527 192.168.29.250 131.253.61.66 TLSv1.2 288 Client Hello
Source Port: 60375
Expand Subtrees Shift+Right
Expand All Ctrl+Right
Collapse All Ctrl+Left
Apply as Column
Apply as Filter
Prepare a Filter
Conversation Filter
Colorize with Filter
Follow
Selected
Not Selected
...and Selected
...or Selected


Wireshark · Follow TCP Stream (tcp.stream eq 35) · wireshark_D32D7F...
.....oSb@N.....b....`Z.y'.....
.B..|..../.....V.....`....jj.+./.,.0...../.5.
...|jj.....login.live.com.....#...
.....h2.http/1.1uP.....
.
...:.....JJ.....U..Y:..ULS.].U:...g5{.....K. .9..P...&.}
b....F..$|...|.....E.....
.....
..
...r0..n0..V.....~.....!tm.x.K..H0
. *.H..
.....0w1.0 ..U....US1.0...U.
..Symantec Corporation1.0...U....Symantec Trust
Network1(0&..U....Symantec Class 3 EV SSL CA - G30..
151120000000Z.
171120235959Z0...1.0...+.....7<.....US1.0...+.....7<....
Washington1.0...U....Private Organization1.0...U... 6004134851.0
7 client pkts, 9 server pkts, 7 turns.
Entire conversation (12 kB)
Show and save data as ASCII
Stream 35
Find:
Find Next
Filter Out This Stream
Print
Save as...
Back
Close
Help




Example 001.pcap Wireshark
Statistics
Summary
Protocol Hierarchy
Conversations
Endpoints
Packet Lengths
I/O Graphs
Conversation List
Endpoint List
Service Response Time
DHCP (BOOTP) Stat...
Compare
Flow Graph...
HTTP
IP Addresses
IP Destinations
IP Protocol Types
OSPF Packet Details
SCTP
UDP Multicast Streams
WLAN Traffic



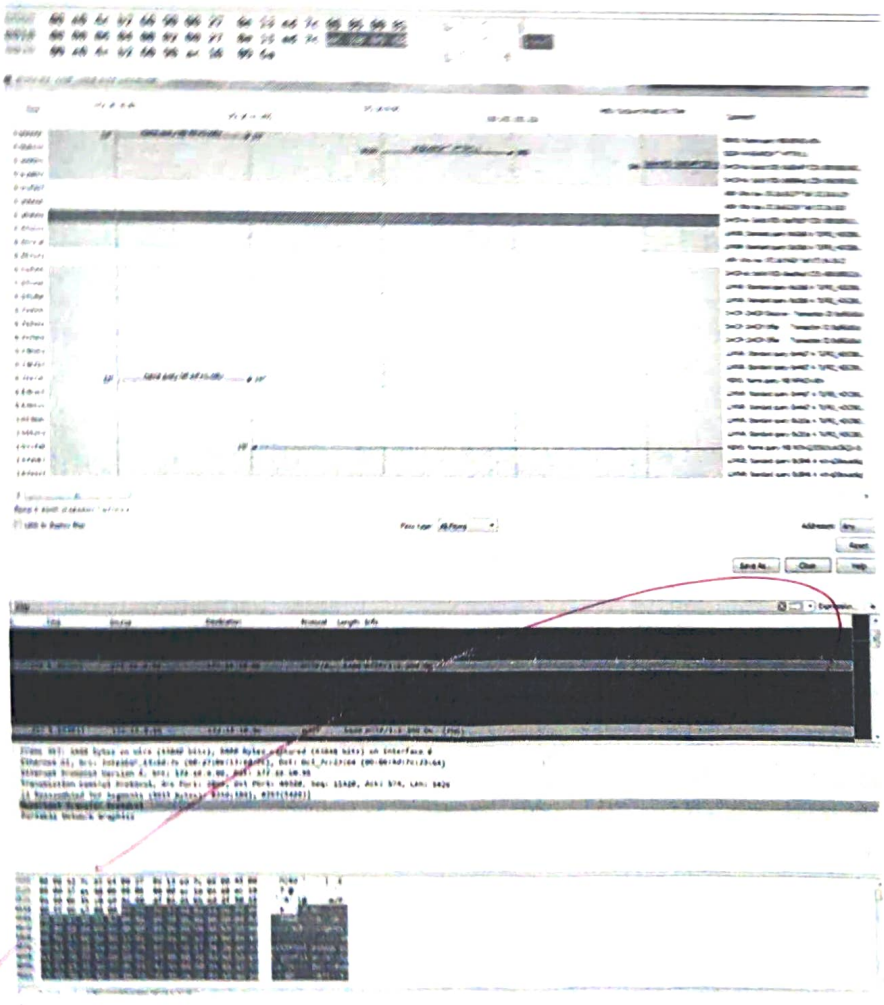
Output

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	Pegatron_e0:87:9e	Broadcast	ARP	60	Who has 172.16.9.94? Tell 172.16.9.138
2	0.000180	RealtekS_55:2c:b8	Broadcast	ARP	60	Who has 172.16.10.36? Tell 172.16.10.50
3	0.000294	RealtekS_55:2c:b8	Broadcast	ARP	60	Who has 172.16.11.36? Tell 172.16.10.50
4	0.000295	RealtekS_55:2c:b8	Broadcast	ARP	60	Who has 172.16.8.37? Tell 172.16.10.50
5	0.000296	RealtekS_55:2c:b8	Broadcast	ARP	60	Who has 172.16.9.37? Tell 172.16.10.50
6	0.000296	RealtekS_55:2c:b8	Broadcast	ARP	60	Who has 172.16.11.37? Tell 172.16.10.50
7	0.001460	fe80::4968:12a7:5e3…	ff02::1:3	LLMNR	95	Standard query 0xae2b A TLFL3-HDC101701
8	0.001622	172.16.8.95	224.0.0.252	LLMNR	75	Standard query 0xae2b A TLFL3-HDC101701
9	0.001623	172.16.8.95	224.0.0.252	LLMNR	75	Standard query 0x28c0 AAAA TLFL3-HDC101701
10	0.001625	fe80::4968:12a7:5e3…	ff02::1:3	LLMNR	95	Standard query 0x28c0 AAAA TLFL3-HDC101701

```
▷ Frame 7: 95 bytes on wire (760 bits), 95 bytes captured (760 bits) on interface 0
▷ Ethernet II, Src: Dell_35:10:a8 (50:9a:4c:35:10:a8), Dst: IPv6mcast_01:00:03 (33:33:00:01:00:03)
◢ User Datagram Protocol, Src Port: 62374, Dst Port: 5355
    Source Port: 62374
    Destination Port: 5355
    Length: 41
    Checksum: 0x90e0 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 0]
▷ Link-local Multicast Name Resolution (query)
```

```
0000  33 33 00 01 00 03 50 9a  4c 35 10 a8 86 dd 60 00   33····P· L5····`·
0010  00 00 00 29 11 01 fe 80  00 00 00 00 00 00 49 68   ···)···· ····Ih
0020  12 a7 5e 36 52 3e ff 02  00 00 00 00 00 00 00 00   ··^6R>·· ········
0030  00 00 00 01 00 03 f3 a6  14 eb 00 29 90 e0 ae 2b   ········ ···)···+
0040  00 00 00 01 00 00 00 00  00 00 0f 54 4c 46 4c 33   ········ ···TLFL3
0050  2d 48 44 43 31 30 31 37  30 31 00 00 01 00 01      -HDC1017 01·····
```





EXNO:6 Error correction at Data link layer

Aim:-

write a program to implement error detection and correction using HAMMING code concept. Make a test run to input data stream and verify error correction feature

Code:-

```
#include <stdio.h>
#include <math.h>
int main()
{
  int data[20], code[20]
  int m, r =0, n;
  int i, j, k=0;
  int errorPOS , syndrome =0;
  Printf ("Enter number of data bits:");
  scanf ("%d", &m);
  Printf ("Enter data bits:");
  for (i=0; i<m : i++);
        scanf ("%d", &data[i]);
  while ((1<<r) < (m+r+1))
    r++;
  n = m+r;
  j=0;
  k=0;
```

```
for (i=0 ; i<=n ; i++)
{ if ( (i & (i-1)) == 0)
        code [i] = 0;

  else
     code [i] = data [j++];
 }
for (i=0 ; i<r ; i++)
  { int parityPos = 1 << i;
    int parity = 0;
  for (j=1 ; j<=n ; j++)
    {
     if ( j & parityPos)
            Parity = Parity ^ code [j];
     }
     code [parityPos] = parity ;
    }
  Printf ("\n generated Hamming code: ");
  for (i=n ; i>=1 ; i--)
          printf ("%d", code [i]);

   scanf ("%d ", & errorPos);
  if ( errorPos != 0)
    {
      code [errorPos] = ! code [errorPos];
  Printf (" Hamming code with error : ")

     for (i=n ; i>= 1 ; i--)
          printf ("%d", code [i]);

   }
```

```
if (Syndrome == 0)
{
  printf ("\n\nNo error detected.");
}
else
{
  Printf ("\n\n Error detected at position
    : %d", syndrome );

      code [syndrome] = !
    code [syndrome];

      printf ("\n corrected Hamming
code :");

      for (i=n; i>=1; i--)
          printf ("%d", code[i]);
    }

    printf ("\n Recovered data :");
    for (i=n; i>=1; i--)
    {
      if ((i & (i-1)) != 0)
          printf ("%d", code [i]);
    }
    Printf ("\n");
    return 0;
  }
```

input :-

1101010

output :-

Error detected at bit position 3

Corrected data : 1111010

Ex No: 7 Flow control at Data link layer

Aim:-

write a program to implement flow control at data link layer using SLIDING WINDOW PROTOCOL - simulate the flow of frame from one node to another

code:-

```
def sliding_window (frames, size):
    total_frame = len (frames)
    sent = 0
    acknowledged = 0
    Print ("In sliding window
PROTOCOL")
    while acknowledged < total frame
    print ("In sender:")
    for i in range (sent, min (sent
window_size, total_frames));
    print ("sending Frame", frame
```

```
print ("\n Receiver:")
for i in range (sent, min (sent + size,
                TOTAL_frames)):
    print ("Frame", frame[i], "received")
    print ("ACK", frame[i], "sent")

acknowledged = min (sent + size,
                TOTAL_frames)
sent = acknowledged
print ("\n window moved. Next
frame:",
        frames[sent] if sent <
TOTAL_frames. else "None")
print ("\n All frame Transmitted
successfully.")
n = int (input ("Enter number of
                frame:"))
frame = []
print ("Enter frame number:")
for i in range (n):
    frames.append (int (input ()))
size = int (input ("Enter size:"))
Sliding_window (frames, size)
```

Output :-

2

sender : sending frame 1

Receiver : Frame 1 received

Receiver : Ack 1 sent

sender : Ack 1 received

sender : sending frame 2

Receiver : Frame 2 received

Receiver : Ack 2 send

sender : Ack 2 received

Exp No:8 NMAP To discover live host, using Nmap scan (ARP, ICMP, TCP/UDP)

Aim:-

To discover Live HOST using Nmap scans (ARP, ICMP, TCP/UDP) on the Try Hack me Platform Room Link: https://tryhackme.com/room/nmap01

Introduction:-

When Targetting a network, we need an efficient tool to handle repetitive Task. This Tool should help us find out which system are active and what service are running on those system. The Tool that we will ~~relying~~ rely on is Nmap.

This room is the first of four in the Nmap series

-> Nmap Live HOST Discovery
-> Nmap basic port scans
-> Nmap Advanced port scans
-> Nmap POST Port scans

we present the different approaches that Nmap uses To discover live hosts.

-> ARP scans: This scans used ARP request To discover ~~live~~ live hosts.

->ICMP Scan: This can use ICMP request To Identify live host

-> TCP/UDP ping scan: scan sends packets To TCP ports and UDP ports to determine live host.

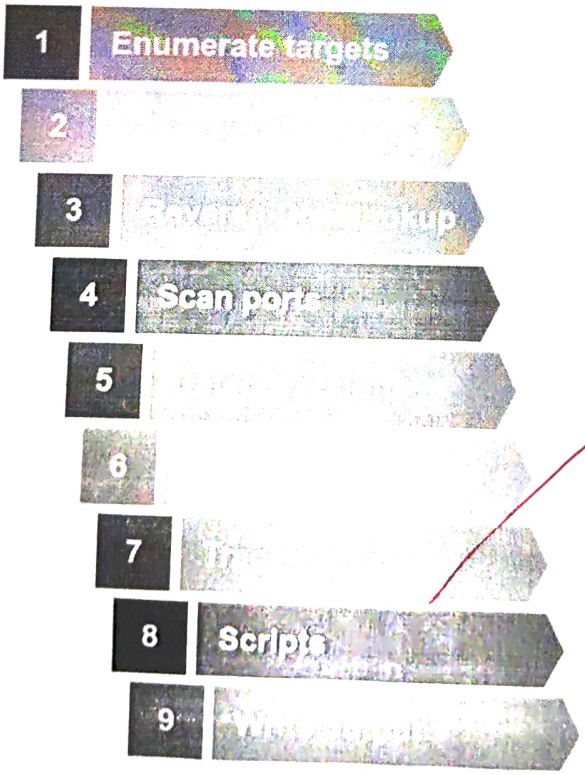
subnetworks:

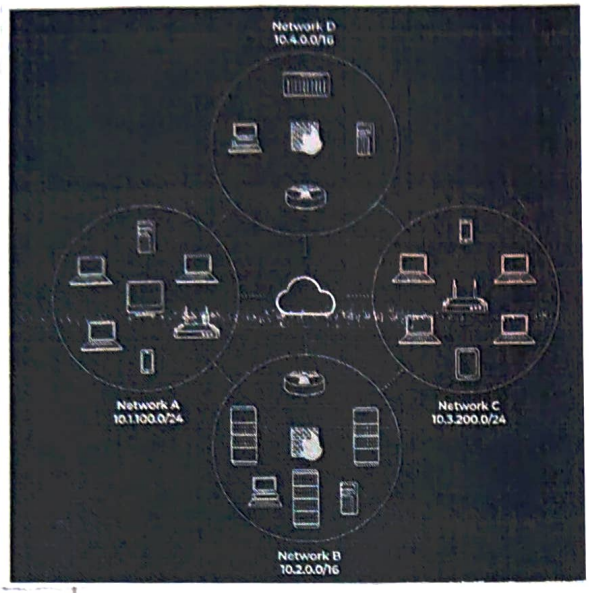
A network segment is a group of computers linked through a shared medium like an Ethernet switch or WIFI access point. In IP network a subnetwork typically consists of one or more network segments connected together and configured to use the same router.

Discovering live hosts

TCP/IP layer :-

→ ARP from link layer
→ ICMP from network layer
→ TCP from Transport layer
→ UDP from Transport layer





Send Packet

From:

computer1

To:

computer1

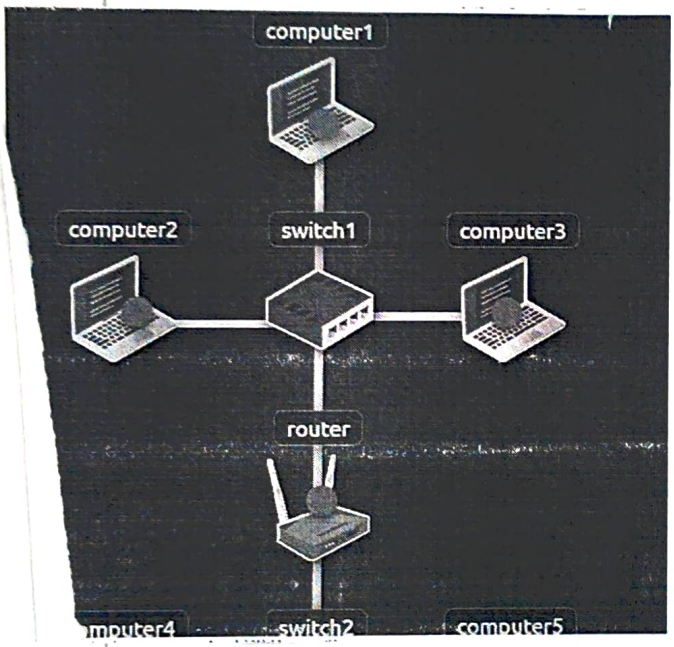
Packet Type:

arp_request

Data:

computer6

Send Packet



Send Packet

From:

computer4

To:

computer4

Packet Type:

arp_request

Data:

computer6

Send Packet

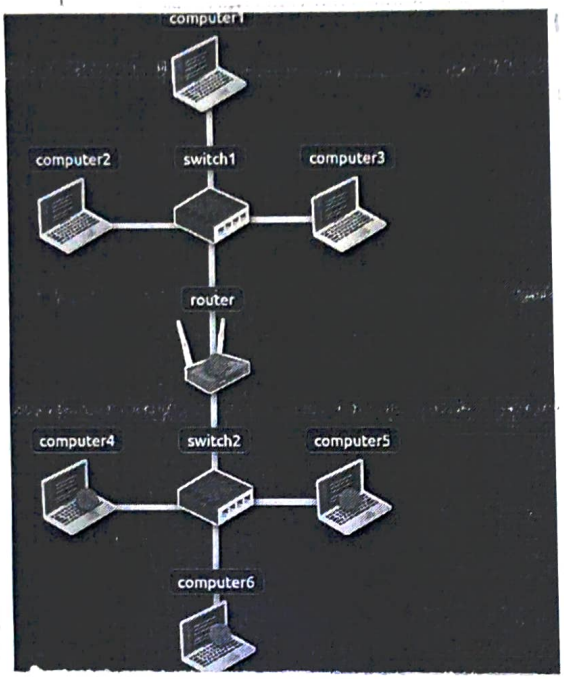
	OSI	TCP/IP	
7	Application Layer	Application Layer	HTTP, HTTPS, SMTP, POP3, IMAP, SSH, FTP, SNMP, Telnet, RDP,...
6	Presentation Layer		
5	Session Layer		
4	Transport Layer	Transport Layer	TCP, UDP
3	Network Layer	Network Layer	IPv4, IPv6, ICMP, IPsec
2	Data Link Layer	Link Layer	ARP, Ethernet (802.3), WiFi (802.11), DSL, Bluetooth,
1	Physical Layer		

→ from computer 4

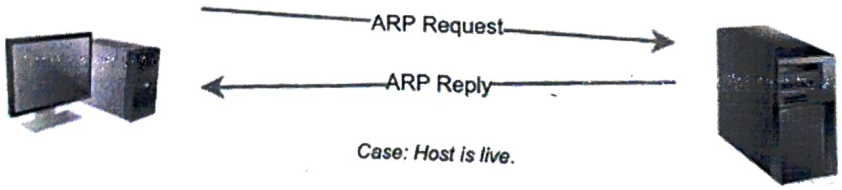
→ To compare 4 (To include its broadcast)

→ Packet Type : ARP Request

→ Data : computer 6



nmap -PR -sn TARGET



Case: Host is live.

nmap -PE -sn TARGET



Case: Host is live.

Nmap uses timestamp request (ICMP Type 13) and checks whether it will get a Timestamp reply (ICMP Type 14). Adding the -PP option tells Nmap to use ICMP timestamp requests. As shown in the figure below, you expect live hosts to reply.

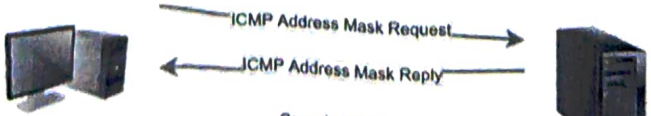
nmap -PP -sn TARGET



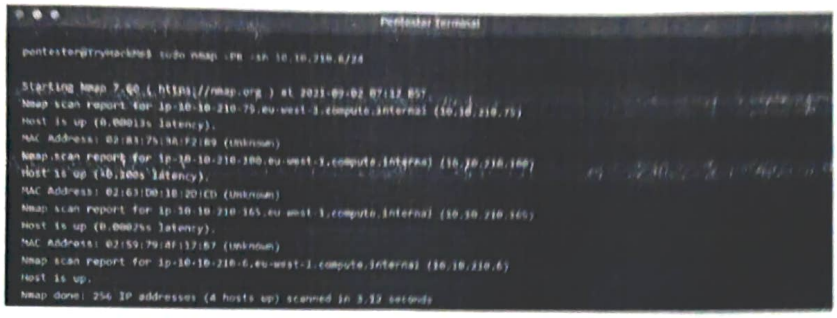
Case: Host is live.

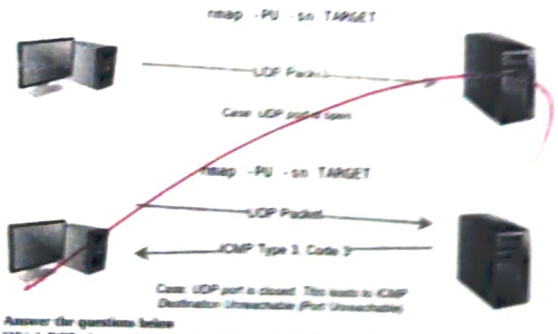
Similarly, Nmap uses address mask queries (ICMP Type 17) and checks whether it gets an address mask reply (ICMP Type 18). This scan can be enabled with the option -PM. As shown in the figure below, live hosts are expected to reply to ICMP address mask requests.

nmap -PM -sn TARGET



Case: Host is live.





Answer the questions below

Which TCP nmap scan does not require a privileged account?

Exp No: 8

8b configuration of wireless LAN using CISCO Packet Tracer

Aim:-
configuration of wireless LAN using CISCO Packet Tracer.

Design a Topology with Three PCs connected from wireless router.

Perform following configuration :-

-> configure Static IP on Pc and wireless Router

-> set SSID To mother network

-> set IP address of router to 192.168.0.1

Pc0 to 192.168.0.2 To 192.168.0.3 .

-> secure your network by configuring WAP key on Router

Step1 : click on wireless router.

-> select Administration Tab from Top menu set username and password To admin and click on save setting.

-> Next click on wireless Tab and set default SSID To mother network

-> Now select wireless security and change security mode to WEP.

->set key 1 To 0123456789

-> Now we have completed all give Task on wireless router Now configure The Static IP on all Three Pc's

-> click an connect Tab and click on referesh

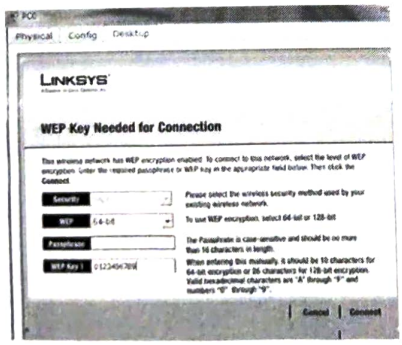
->IT will ask for WAP key insert 0123456789 and click connect

->IT will connect you with wireless router

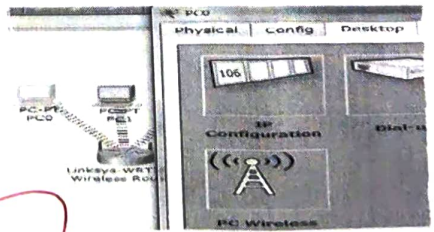
Student observation :-

a) what is SSID of a wireless Router?

SSID (service set identifier) is The name a wireless network (wi-Fi). IT is used To identify and connect devices To particular wireless network.



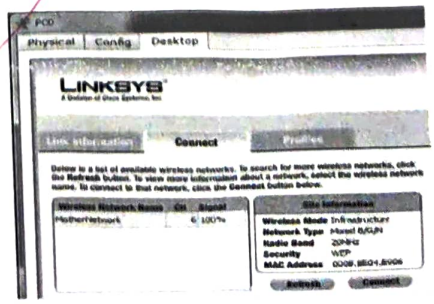
connect you with wireless router



u can see in image below that system is connected. And PCI card is



lick on connect tab and click on Refresh button



Pg:9

Implementation of subnetting in CISCO Packet Tracer.

Aim:

Implementation of subnetting in CISCO Packet Tracer simulator

creating A network Topology:

The first step in implementing classless IP subnetting is to create a network topology in packet Tracer. To create a network topology in packet tracer, select the "new" button in the top left corner, then select "Networks" and "generic".

Subnetting :-

To subnet the network address of 192.168.1.0/24 to provide enough space for at least 5 addrers for end devices, the switch, and the router, we can use a /27 subnet mask.

The IP addressing for the network shown in the Topology can be as follow.

-> Router R1:

-> Gigabit Ethernet 0/0 : 192.168.1.1

-> Gigabit Ethernet 0/1 : 192.168.2.1

->switch S1 :

-> Fast Ethernet 0/1 : 192.168 .1.0/27

-> Pc 1 : 192.168.1.11

-> Pc 2 : 192.168.1.12

-> PC3 : 192.168.1.13

-> PC4 : 192.168.1.14

-> PC5 : 192.168.1.15

-> Fast Ethernet 0/2 : 192.168.2.0/27

-> PC1 : 192.168.2.11

-> PC2 : 192.168.2.12

-> PC3 : 192.168.2.13

-> PC4 : 192.168.2.14

-> PC5 : 192.168.2.15

-> Router R2:-

-> Fast Ethernet 0/0 : 192.168.3.1

-> Fast Ethernet 0/1 : 192.168.4.1

-> Switch S2:-

-> Fast Ethernet 0/1 : 192.168.3.0/27

-> PC1 : 192.168.3.11

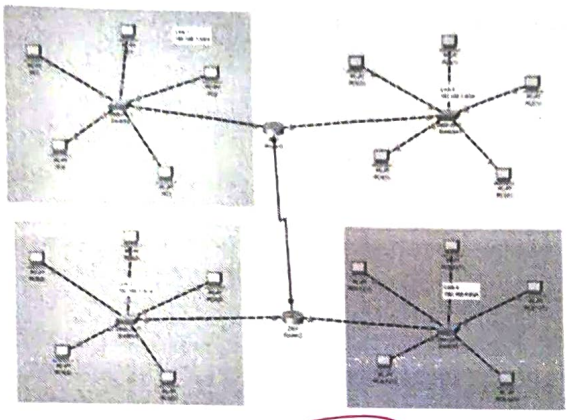
-> PC2 : 192.168.3.12

-> PC2 : 192.168.3.13

-> PC2 : 192.168.3.14

Testing The Network :-

Now that our network topology is configured, we can test the network open a command prompt on each PC and try toping the other PC. If the ping is succesfuley, then the network is functioning properly. we can also use the "ping" command to test connectivity between the router and the PCs



Exp No: 10 Internet working with routers in CISCO Packet Tracer simulator

Aim :-

internet working with routers in CISCO packet Tracer simulator

Design and configure a simple inter network using a router

In this network, a router and 2 PCs are used computer are connected with router using a copper straight - through cable after forming the network to check network connectivity a simple from PC0 TO PC1

Procedure :-

Step 1 :-

-> select the router and open CLI

-> press ENTER TO start configuring Router 1

-> Type enable TO activate the priveleged mode

Step 2 :-

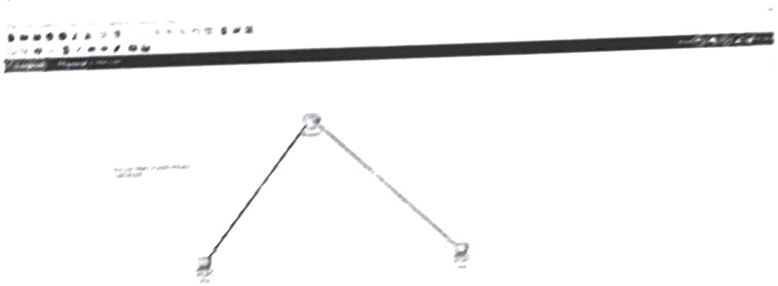
-> connect Fast Ethernet Port of PC0 with fast Ethernet Port of Router 0 using a copper straight - Through cable

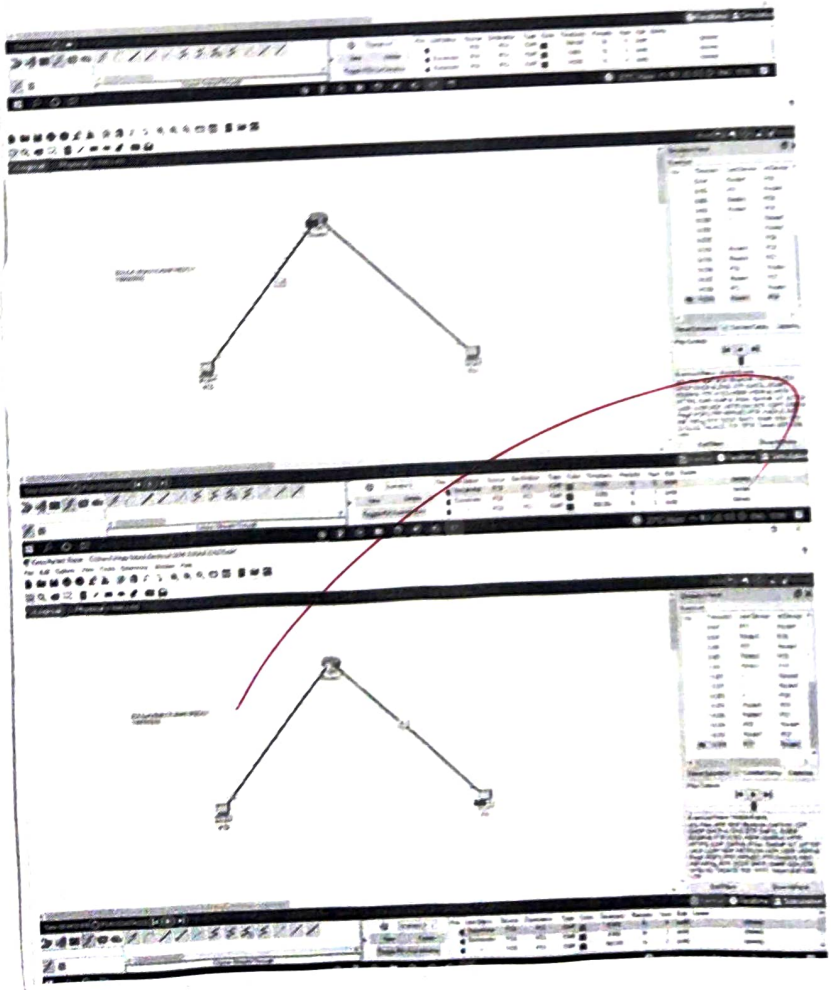
Router configuration Table :-

Device name	IPaddress FastEthernet 0/0	Subnetmask	IPaddress Fast Ethernet 1/0
Router	192.168.0.1	255.255.255.0	192.168.20.1

Pc configuration Table :-

Device name	IPaddress	subnet mask	gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
Pc1	192.168.20.2	255.255.255.0	192.168.20.1





Part 2:- configure the Network Device

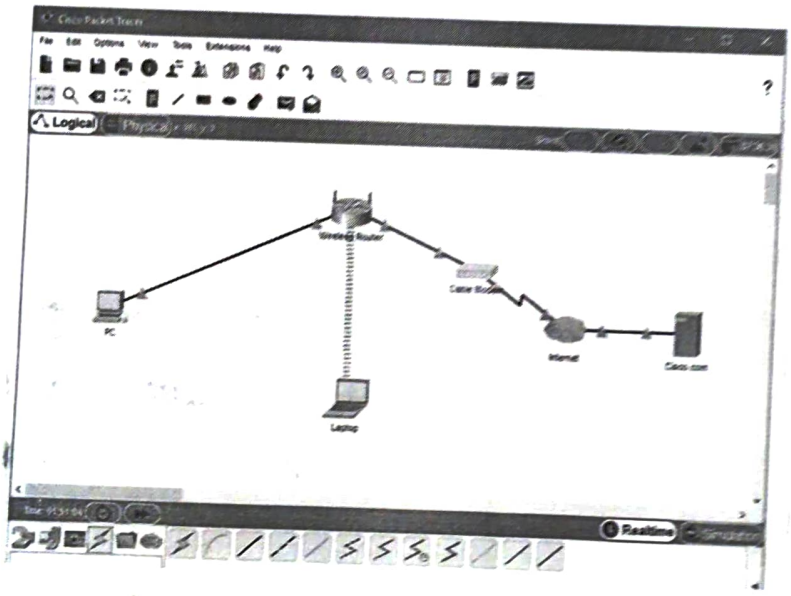
Step 1: configure the wireless router

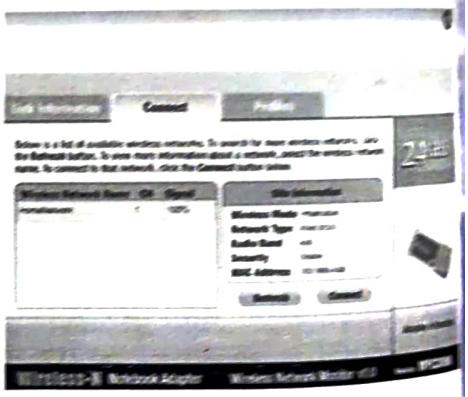
a) create the wireless network on the wireless router

b) click on the save setting tab

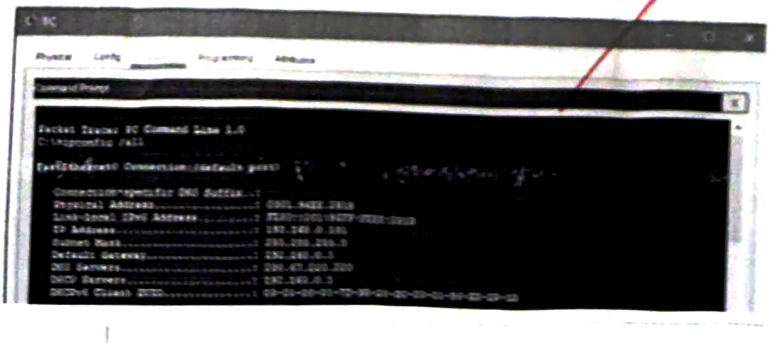
Step 2:- configure the laptop

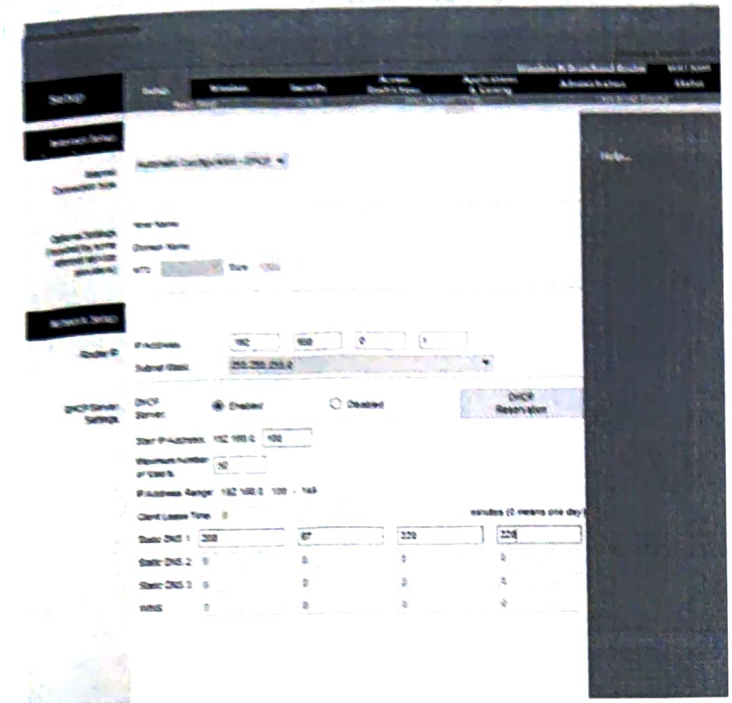
a) configure the Laptop to access the wireless network

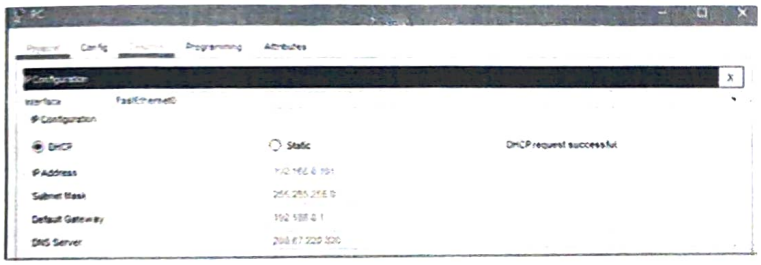




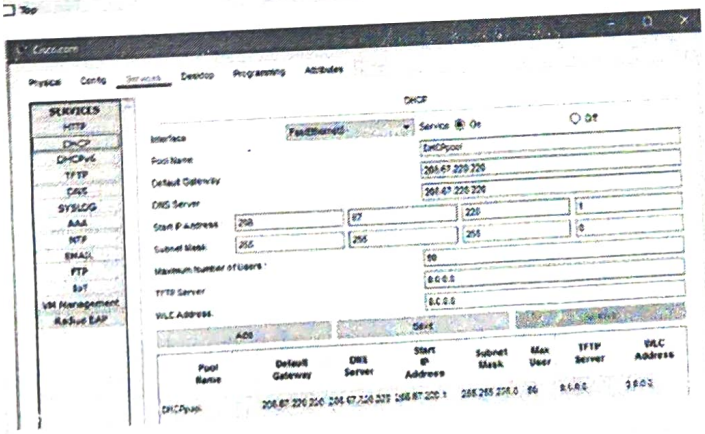




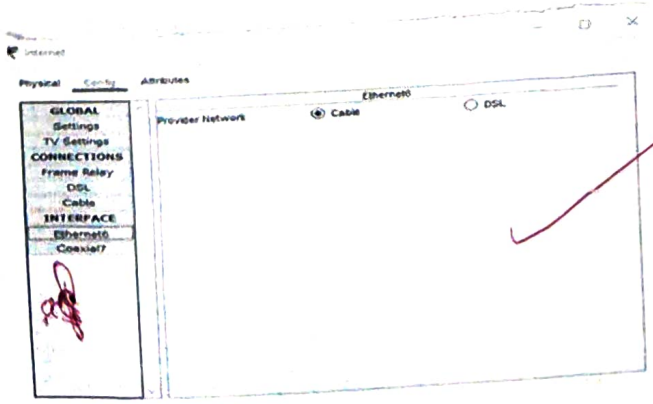




Click on the Command Prompt icon. Verify that the PC has received an IPv4 address by issuing the **ipconfig /all** command from the command prompt as shown in the figure. The PC should receive an IPv4 address in the 192.168.0.x range.



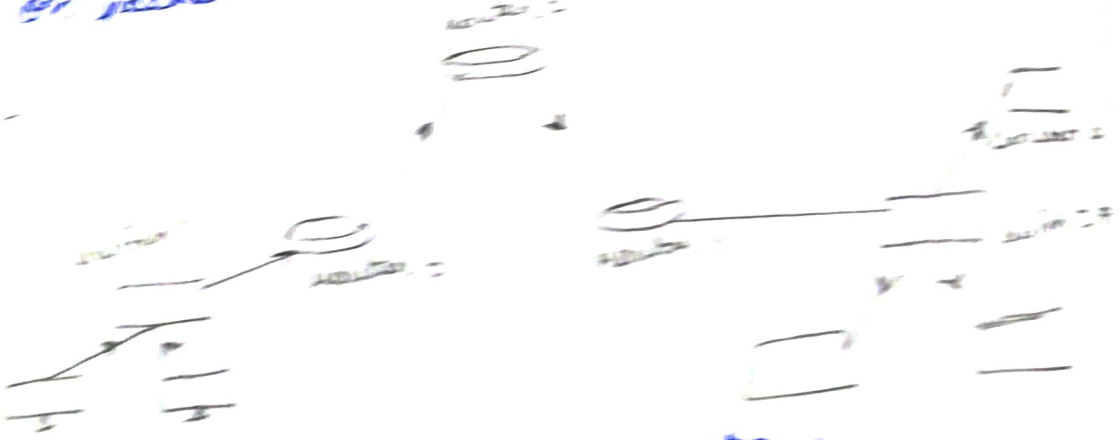




Expt:- Simulate static routing configuration using cisco packet tracer

Aim:-
Simulate static routing configuration using cisco packet tracer

Setting up lab:-
Here's a packet trace lab we create in the following image or download the following and load it for packet tracer with initial IP configuration



Creating, adding, modifying static routes:-
Static routes, sometimes known also connected routes, are only need to add routes for the network that are not available on the router's interface. For example network 10.0.1.0/8, 20.20.20.0/8 and 30.0.0.0/8 are directly connected

Router 1 requirements:-
-> Create two routes to network 10.0.0.0/8 and configure the first route as the main route and the second route will be backup the router.
-> Verify the routes and only main routes on the routing table

Router 2 requirements:-
Create static routes to network 10.0.0.0/8 and network 30.0.0.0/8 and verify the routes added now

router To The routing Table

Verifying static routing

on router 1, we configured two router for network 30.0.0.0/8. These route are via router 1 and via router 2, we set the first router as the main router and the second router as the backup.

Deleting a static router:

-> use the show up route static, command to print all static router

-> Note down the router you want to delete.

-> use the no ip router command To delete the router.

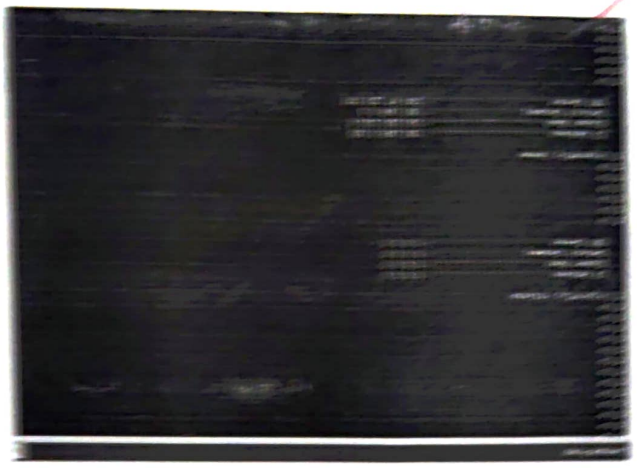
Router 1 (config)# router

Router 1 (config-router)# network 10.0.0.0

router 1 (config-router)# network 192.168.1.252

=> router ip command till router To enable the ~~RIP~~ RIP routing protocol

Network command allows us To specify The Network which we want To advertise.



10: 12a) End - End communication at Transport Layer

Aim:-
Implement echo client using TCP / UDP sockets.

Algorithm:-

```
import socket
import Threading
def start_server():
    server = socket.socket(socket.AF_INET, socket_stream)
    server.bind(('127.0.0.1', 5000))
    server.listen(1).
    print("[server] listening on 127.0.0.1.5000")
    conn, addr = server.accept()
    print(f"[server] connected by [addr]")
while True:
        data = conn.recv(1024).decode()
        if not data or data.lower() == 'quiet'
          print('[server] connection closed by client')
        break
        print(f"|server) received: {data}")
 conn.close()
server.close()
if __name__ == "__main__":
```

output:-

[server] listening on 127.0.0.1.5000 [client] connected

server

[server] connected by ('127.0.0.1', 53458) Enter message (or quit to exit):

hello

[server] Received: hello

[server Response] server received: hello

Ex: 12b End-End communication at Transport Layer

Aim:-
Implement chat client server using TCP / UDP packet sockets.

Algorithm:-

```
import socket
import Threading
def udp_server():
s= socket.socket (socket.AF_INET, socket.sock_Dgra
s.bind ("[server]running on 127.0.0.1:12000"
while True:
data, addr = s.recfrom (1024)
msg = data.decode()
if msg.lower() == 'quit':
print (f"[server] client {addr} ended
The connection.")
break
print (f"[server] received from {addr}: {msg}")
s.sendto (data, addr)
s.close()
print ("[server] stopped")
def udp_client():
c = socket.socket (socket, AF_INET, socket.
sock_DGRAM)
server = ('127.0.0.1', 12000)
while True:
msg = input ("Enter message (or quit to stop):")
c.sendto (msg.encode(), server)
s.close()
print ("[server] stopped")
def udp_client():
```

while True :-

```
msg = input ("Enter message (or 'quit' to exit): ")
c.sendto (msg.encode(), server)
if msg.lower() == "quit":
    print ("[Client] closing connection...")
    break
data, = c.recvfrom (1024)
print ("Echo from server:", data.decode())
c.close()
udp.client()
```

Output:

```
Enter message (or 'quit' to exit): [server]
running on 127.0.0.1:1200
[server] Received from ('127.0.0.1', 49670): hi
Echo from server: hi
Enter message (or 'quit' to exit): hello.
```

EXP NO: 13

Implement your own ping program

Aim:-

Implement your own ping program

Algorithm:-

```
import os
import platform
import subprocess

def ping(host = "google.com", count=4):
    = "-n" if platform.system().lower() ==
        "windows" else "-c".
command = ["ping", param, str(count), host]

Try:
output = subprocess.check_output(command, universal
                                 newlines = True)

    print(output)
if-name == "__main":
    ping("8.8.8.8")
```

Output:-

Pinging 8.8.8.8 with 32 bytes of data:

reply from 8.8.8.8 bytes = 32 times = 6ms TTL = 119

Ping statistics for 8.8.8.8

Packets: sent = 4, received = 4, lost = 0 (0% loss)

Approximate round trip time in milli-seconds

Minimum = 5ms, maximum = 9ms, Average = 7ms.

EXP NO: 14

RAW socket to implement packet sniffing

Aim:

to determine the RAW sockets to implement packet sniffing

code:

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP

def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst
        protol_name = ""
        if protocol == 1:
            protocol_name = "ICMP"
        elif protocol == 6:
            protocol_name = "TCP"
        elif protocol == 17:
            protocol_name = "UDP"
    else:
        protocol_name = "unknown Protocol"
    print(f"protocol: {protocol_name}")
    print(f"source IP: {src_ip}")
    print("-" * 50)
sniff(iface = 'wifi', prn = packet_callback,
      filter = "ip", store = 0)
```

EXP NO: 14

RAW socket to implement packet sniffing

Aim:-

to determine the RAW sockets to implement packet sniffing

code:-

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP

def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst
        protol_name = ""
        if protocol == 1:
            protocol_name = "ICMP"
        elif protocol == 6:
            protocol = name = "TCP"
        elif protocol == 17:
            protocol_name = "UDP"
else:
    protocol_name = "unknown Protocol"
    print(f"protocol: {protocol_name}")
    print(f"source IP: {src: ip}")
    print("-"*50)
sniff(iface = 'wifi', = packet_callback,
      filter = "ip", store = 0)
```

Internet :-

Protocol : TCP

Source IP : 54.155.141.117

Destination IP : 192.168.1.6

Protocol : UDP

Source IP : 192.168.1.6

Destination IP : 192.168.1.1

(completed)

13/8/26